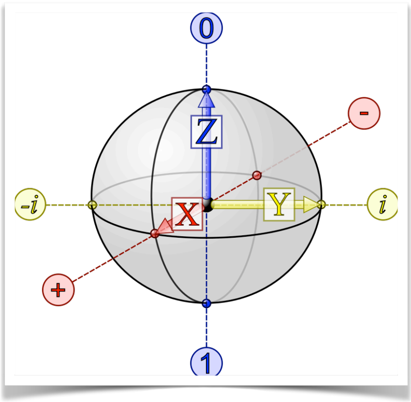

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IQC 2024/25 (756AA, 6CFU)

Introduction to Quantum Computing

Superdense Coding

Superdense coding

Simple, yet surprising application of elementary quantum mechanics
It involves two parties, **Alice** and **Bob**, who are far away from one another

GOAL

Alice is in possession of **two classical bits** *of information, which she wishes to send to Bob* **USING A SINGLE QUBIT**

This protocol was

- first proposed by Charles H. Bennett and Stephen Wiesner in 1970 (though not published by them until 1992)
- and experimentally realized in 1996 using entangled photon pairs

Superdense coding can be thought of as the opposite of quantum teleportation, in which one transfers one qubit from Alice to Bob by communicating two classical bits

Discoverers of superdense coding

Charles H. Bennett

Stephen Wiesner

Superdense coding

Quantum communication protocol

- **Quantum communication channel**

  refer to a communication line (e.g., a fiber-optic cable), which can carry qubits between two remote location

- **Communication channel**

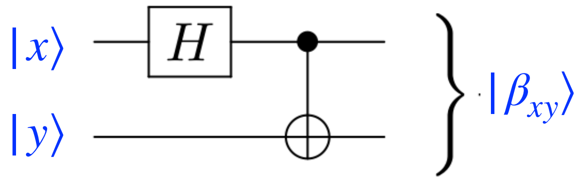
  refer to a channel which can carry classical bits (but not qubits)

Alice and Bob initially share a pair of qubits in an entagled state, **one qubit in Alice's possession, and the other in Bob's possession**

Bell state

$$|\beta_{00}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Bell states



$$|00\rangle \xrightarrow{H\otimes I} \frac{|0\rangle + |1\rangle}{\sqrt{2}}|0\rangle \xrightarrow{CNOT} \frac{|00\rangle + |11\rangle}{\sqrt{2}} = |\beta_{00}\rangle$$

$$|01\rangle \xrightarrow{H\otimes I} \frac{|0\rangle + |1\rangle}{\sqrt{2}}|1\rangle \xrightarrow{CNOT} \frac{|01\rangle + |10\rangle}{\sqrt{2}} = |\beta_{01}\rangle$$

$$|10\rangle \xrightarrow{H\otimes I} \frac{|0\rangle - |1\rangle}{\sqrt{2}}|0\rangle \xrightarrow{CNOT} \frac{|00\rangle - |11\rangle}{\sqrt{2}} = |\beta_{10}\rangle$$

$$|11\rangle \xrightarrow{H\otimes I} \frac{|0\rangle - |1\rangle}{\sqrt{2}}|1\rangle \xrightarrow{CNOT} \frac{|01\rangle - |10\rangle}{\sqrt{2}} = |\beta_{11}\rangle$$

EVERY VECTOR
IN THIS BASIS IS
ENTANGLED

How to share a pair of entangled qubits?

- The state $|\beta_{00}\rangle$ would have to be created ahead of time, when the qubits are in a lab together and can be made to interact in a way that will give rise to the entanglement between them

- After the state is created, Alice and Bob each take one of the two qubits away with them

ALTERNATIVELY

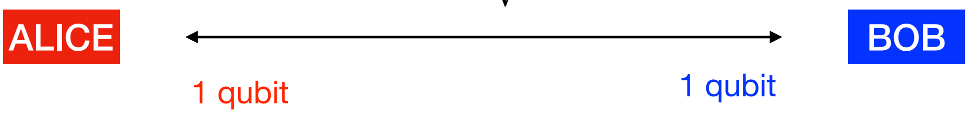
- Some third party may prepare the entangled state ahead of time, sending one qubit to Alice and the other to Bob

- If they are careful not to let them interact with the environment, or other quantum systems, Alice and Bob's joint state will remain entangled

How to share a pair of entangled qubits?

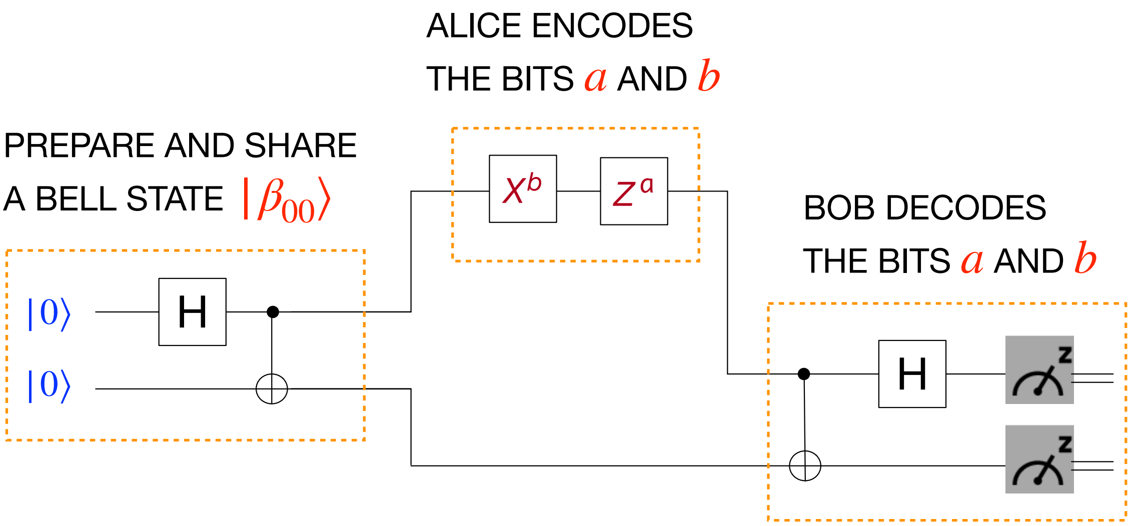
Note that $|\beta_{00}\rangle$ is a fixed state, there is no need for Alice to have sent any qubit to Bob to prepare this state

$$|\beta_{00}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$



We will show how Alice can send two bits of information to Bob, sending the single qubit in her possession

Quantum circuit for superdense coding



$$H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \qquad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \qquad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$X^1 = X \qquad Z^1 = Z \qquad X^0 = Z^0 = I$$

The procedure

$$|\beta_{00}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

To send $a\,b$	$X^b \;\; Z^a$	Resulting state
00	I	$|\beta_{00}\rangle \xrightarrow{I\otimes I} |\beta_{00}\rangle$
01	X	$|\beta_{00}\rangle \xrightarrow{X\otimes I} \frac{|10\rangle + |01\rangle}{\sqrt{2}} = |\beta_{01}\rangle$
10	Z	$|\beta_{00}\rangle \xrightarrow{Z\otimes I} \frac{|00\rangle - |11\rangle}{\sqrt{2}} = |\beta_{10}\rangle$
11	ZX	$|\beta_{00}\rangle \xrightarrow{X\otimes I} \frac{|10\rangle + |01\rangle}{\sqrt{2}} \xrightarrow{Z\otimes I} \frac{-|10\rangle + |01\rangle}{\sqrt{2}} = |\beta_{11}\rangle$

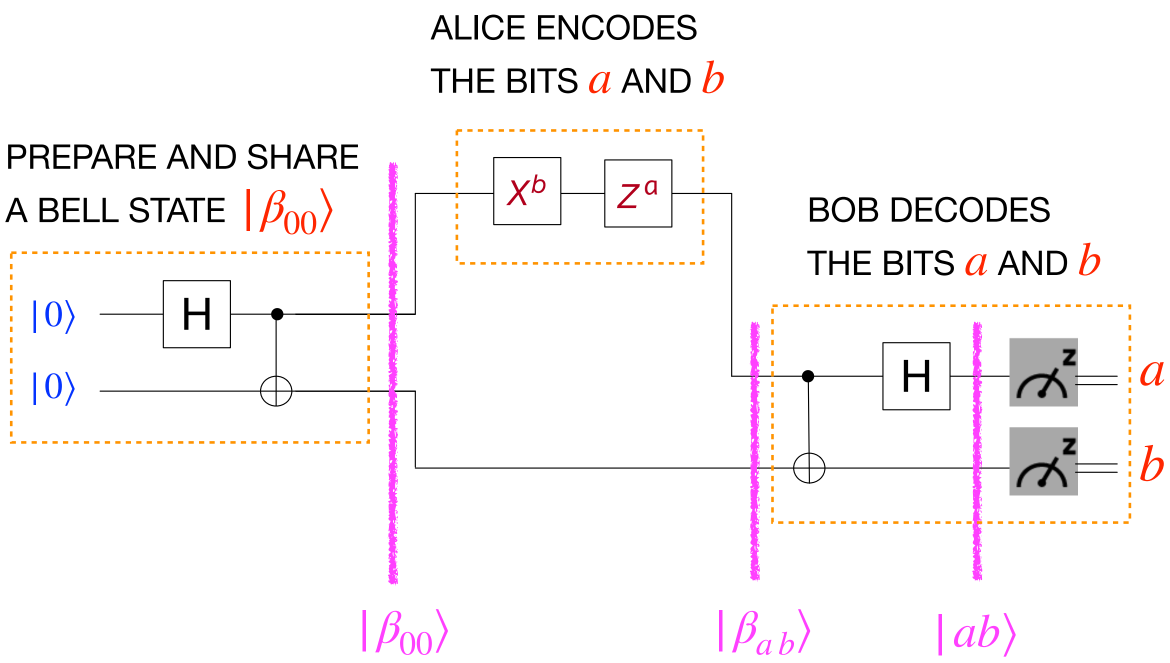
The procedure

The resulting state state $|\beta_{a\,b}\rangle$ is one of the four Bell states

Bell states form an orthonormal basis, and can be distinghished by an appropriate quantum measurement (in the Bell basis)

- After applying the appropriate gate/gates, Alice sends her qubit to Bob
- Bob is in possession of one of the four Bell states, depending on the classical bits that Alice wished to send him
- Bob performs a measurement in the Bell basis and determines which bit pair Alice sent
- The measurement in the Bell basis can be implemented by first performing a change of basis (Bell → computational), and then performing a measurement in the computational basis
- The outcome of the Bell measurement reveals to Bob the Bell states he possesses, and allows him to determine **with certainty** the two classical bits

Quantum circuit for superdense coding



The measurement gives Bob the values $a,\ b$ corresponding to the Bell state $|\beta_{ab}\rangle$ in his possession, with certainty

Example

- Alice wants to send the bits $10\ (a = 1,\ b = 0)$
- Since $X^b Z^a = X^0 Z^1 = IZ = Z$, Alice applies the Pauli gates Z to her qubit of the initial shared Bell state $|\beta_{00}\rangle$

$$(Z \otimes I)|\beta_{00}\rangle = (Z \otimes I)\left(\frac{|00\rangle + |11\rangle}{\sqrt{2}}\right) = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

- Then, Alice sends her qubit to Bob
- Bob performs the measurement: he applies a CNOT with the first qubit as control, and the second as target, and then a Hadamard on the first qubit:

$$\frac{|00\rangle - |11\rangle}{\sqrt{2}} \xrightarrow{CNOT} \frac{|00\rangle - |10\rangle}{\sqrt{2}} \xrightarrow{H \otimes I} \frac{|+\rangle|0\rangle - |-\rangle|0\rangle}{\sqrt{2}}$$
$$= \frac{1}{\sqrt{2}}\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}|0\rangle - \frac{|0\rangle - |1\rangle}{\sqrt{2}}|0\rangle\right)$$
$$= \frac{1}{2}(\cancel{|00\rangle} + |10\rangle - \cancel{|00\rangle} + |10\rangle) = |10\rangle$$

- Now Bob has the basis state $|10\rangle$ and knows that Alice sent 10

Discussion

- Alice, interacting with only a single qubit, is able to transmit two bits of information to Bob
- Two qubits are involved in the protocol, but Alice never needs to interact with the second qubit
- Classically, the task Alice accomplishes would have been impossible had she only transmitted a single classical bit

Information is physical, and surprising physical theories as Quantum Mechanics may predict surprising information precessing abilities